INPUT / USB-C PD negotiation

RSET = 56k requests fixed 15 V PDO. A 35 W label alone does not guarantee 15 V.

J1

USB-C PD 15V/3A

F1

3A 63VDC FAST

DTVS

SMAJ16A

UPD

CH224A

RSET

56k 1%

CPD

1uF 50V X5R

CSYS

1uF 50V X5R

PG intentionally unused; PCB single-pad /PD_GOOD has no copper.

PWR_FLAG: external source/GND, fuse output, and LC-derived rails.

Power flags document sources behind passive parts; they add no hardware.

INPUT / eFuse and system enable

UIN

TPS2663DRGER

RUVINH

100k 0.1%

RUVINL

10k 0.1%

ROVINH

133k 0.1%

ROVINL

10k 0.1%

MODE / SHDN open: latch-off fault mode / internally enabled.

NC19-24 are unused. EP25 and GND8 both connect to GND.

TPIN

INPUT FAULT

TPPG

INPUT PGOOD

Common ground. Independent regulators and protection; no galvanic isolation.

PROTECTED INPUT / shared polymer bank

RPGH

100k 0.1%

RPGL

10k 0.1%

RPGPU

100k

CBULK1

82uF 35V POLYMER

CBULK2

82uF 35V POLYMER

CBULK3

82uF 35V POLYMER

82uF x3 / 35V. Keep local MLCCs at each buck VIN.

Design target: 5 V / 1.5 A continuous per channel.

Verify load transient, ripple and temperature on first article.

RILIMIN

8.25k 1%

CDIN

100nF 50V

H1

M2 NPTH / KEEP DIA7

H2

M2 NPTH / KEEP DIA7

LEFT / fixed 4.984 V buck

VOUT = 0.8 x (1 + 52.3k / 10k). RT7272B + 6.8uH.

U2

RT7272BGSP

CIN2A

10uF 50V X5R

CIN2B

10uF 50V X5R

CB2

100nF 50V

L2

6.8uH / 6A Isat

COUT2A

22uF 25V X5R

COUT2B

22uF 25V X5R

RFH2

52.3k 0.1%

RFL2

10k 0.1%

RC2

24k

CC2

2.7nF C0G 50V

RLIM2

84.5k 1%

CE2

100uF 25V POLYMER

CUSB2

1uF 50V X5R

CENTRE / fixed 4.984 V buck

VOUT = 0.8 x (1 + 52.3k / 10k). RT7272B + 6.8uH.

U3

RT7272BGSP

CIN3A

10uF 50V X5R

CIN3B

10uF 50V X5R

CB3

100nF 50V

L3

6.8uH / 6A Isat

COUT3A

22uF 25V X5R

COUT3B

22uF 25V X5R

RFH3

52.3k 0.1%

RFL3

10k 0.1%

RC3

24k

CC3

2.7nF C0G 50V

RLIM3

84.5k 1%

CE3

100uF 25V POLYMER

CUSB3

1uF 50V X5R

RIGHT / fixed 4.984 V buck

VOUT = 0.8 x (1 + 52.3k / 10k). RT7272B + 6.8uH.

U4

RT7272BGSP

CIN4A

10uF 50V X5R

CIN4B

10uF 50V X5R

CB4

100nF 50V

L4

6.8uH / 6A Isat

COUT4A

22uF 25V X5R

COUT4B

22uF 25V X5R

RFH4

52.3k 0.1%

RFL4

10k 0.1%

RC4

24k

CC4

2.7nF C0G 50V

RLIM4

84.5k 1%

CE4

100uF 25V POLYMER

CUSB4

1uF 50V X5R

OUTPUT / UVLO, OVLO and current limiting

UP2

TPS259470LRPWR

RUVH2

28k 0.1%

RUVL2

10k 0.1%

ROVH2

34k 0.1%

ROVL2

10k 0.1%

J2

XH2.50-2P 5V OUTPUT

TP2

FAULT LEFT

CD2

8.2nF C0G 50V

RILM2

1.65k 1%

OUTPUT / UVLO, OVLO and current limiting

UP3

TPS259470LRPWR

RUVH3

28k 0.1%

RUVL3

10k 0.1%

ROVH3

34k 0.1%

ROVL3

10k 0.1%

J3

XH2.50-2P 5V OUTPUT

TP3

FAULT CENTRE

CD3

8.2nF C0G 50V

RILM3

1.65k 1%

OUTPUT / UVLO, OVLO and current limiting

UP4

TPS259470LRPWR

RUVH4

28k 0.1%

RUVL4

10k 0.1%

ROVH4

34k 0.1%

ROVL4

10k 0.1%

J4

XH2.50-2P 5V OUTPUT

TP4

FAULT RIGHT

CD4

8.2nF C0G 50V

RILM4

1.65k 1%

OBD Three Gauge / Electrical schematic

Sheet: /

File: power_pd_3x5v_r3.xh.kicad_sch

Title: USB-C PD 15V / 3x5V Protected Power

Size: A2	Date: 2026-10-26	**Rev: R3-ERC-20261026**
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